# The Book of Inquiry

## A Handbook for Human–AI Discovery

# Preface

This handbook documents the evolution from using AI as a question-answering tool to treating it as a partner in disciplined inquiry.

# Part I – The Journey

We began by searching for better prompts. We ended by designing a research methodology. Each stage corrected weaknesses in the previous stage instead of merely adding complexity.

# The Seven Stages

1. Hidden project architecture.
2. Opportunity discovery.
3. Falsification.
4. Civilization-level reasoning.
5. Reality audit.
6. Discovery engine.
7. Augmented epistemology.

# Core Lessons

• Better questions outperform better answers.
• Elegant theories are not necessarily true.
• Reality is the final arbiter.
• Mechanisms matter more than patterns.
• Unknowns are valuable outputs.
• Confidence must follow evidence.
• Every theory should compete with alternatives.
• A methodology compounds; a single idea usually does not.

# Research Operating System

Observe → Evidence → Ontology → Assumptions → Residuals → Representation → Competing theories → Mechanism → Constraints → Predictions → Falsification → Revision.

# Universal Mental Models

Residuals, Constraints, Feedback, Incentives, Representation, Selection, Emergence, Networks, Attention Allocation, Information, Optimization.

# Historical Benchmark Program

Rediscover Darwin, Einstein, Shannon, Turing, Wegener, Semmelweis and others using only evidence available before their breakthroughs. Measure anomaly detection, mechanisms, predictions and false positives.

# Prediction Program

Maintain a prediction journal. Every prediction must be specific, dated and falsifiable. Review every 6, 12 and 36 months.

# Open Questions

Why is reality compressible? Why do abstractions work? How should finite intelligence allocate attention? Can discovery itself be improved?

# Quotes & Axioms

'Reality outranks elegance.'
'Compression without prediction is philosophy.'
'A pattern without a mechanism is a question pretending to be an answer.'
'The quality of a mind is determined less by the answers it reaches than by the process it is willing to subject those answers to.'
'The next question is always more valuable than the last certainty.'

# Future Roadmap

Expand into domain-specific playbooks (science, business, software, economics), maintain failed theories, benchmark predictions annually, and continuously revise the methodology.